Hasibur Rahman

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building and supporting production software, including **Golang microservices, REST APIs, event-driven systems, data pipelines, and AI-enabled platforms**. **Proven track record leading 3–4 engineer teams** and working across APIs, databases, messaging, caching, cloud infrastructure, observability, and production debugging.

EXPERIENCE

VivaSoft Ltd June 2024 – Present
Software Engineer Dhaka, Bangladesh

- **Currently leading Kothon Analytics**, a production **Voice AI + Business Intelligence platform** that processes call-center conversations into actionable business insights, profiling, issue detection, escalation signals, and owner-level analytics processing over **125+ hours of calls daily**
- **Led 4-engineer team architecting 6 microservices** with concurrency-safe designs, fault tolerance, structured logging, and production observability
- Designed and maintained production **REST APIs** and backend services deployed on GCP (Cloud Run, Redis, GCS) using Docker and Terraform
- Architected a **Fact/Event Store** backed by an asynchronous **RabbitMQ** pipeline with dead-letter recovery and an **exactly-once dispatch ledger** (idempotency-key claim pattern via a single UNIQUE constraint) for reliable backend data ingestion and event delivery
- Architected a granular **EAV data model** to decouple entities from their attributes and values, allowing new insight types and evolving business attributes to be added without database schema changes
- Designed **multi-tenant data isolation** across call-linked and definition-table schemas with distinct, enforced scoping rules per table family, supporting per-department/business-unit segmentation across many client companies on one platform
- Investigated and traced a production **RabbitMQ acknowledgement bug** where messages were acknowledged immediately after delivery into a Go channel buffer; identified that messages appearing absent from RabbitMQ were actually accumulating in **process memory**, isolating the channel buffer as the source of the misleading queue state
- Designed and architected a call **state machine for granular lifecycle tracking and event decoupling**, enabling extensible, state-aware analytics and downstream features such as per-call insight generation
- Designed **Redis-backed** referential-integrity safeguards protecting critical backend and AI data pipelines from silent corruption
- Implemented production monitoring and observability using **Prometheus and Grafana**, supporting investigation of backend service behavior and operational issues
- Engineered a production-grade **MCP (Model Context Protocol) server** with **header-based authentication**, interoperable with **OpenCode, Cline**, and other MCP-compliant AI clients — exposing internal analytics and backend capabilities as agent-callable tools
- Led **AI based Business Intelligence Platform** for business owners integrated with call centers, connecting backend services, analytics, and AI-powered workflows
- Integrated **MAKER** mechanism to ensure reliable agentic decision **tested on 1M step task**
- Mentored junior engineers across backend service development, STT integration, and data platform development
- **Tech Stack:** Golang, FastAPI, React, Redis, RabbitMQ, MCP (header-based auth), OpenCode, GCP, GCS, Google Artifact Registry, Docker, Terraform, Prometheus, Grafana, Microservices, LangChain, Gemini, MySQL

Chaldal (YC-backed e-commerce, 1.2M users) March 2023 – June 2024
Software Engineer Dhaka, Bangladesh

- Built fraud prevention logic **reducing loss by 40tk/order** using efficient, **sargable database queries** and clean backend code
- Developed address change flow **used by 1M+ users**; **improved saving efficiency by 72%**
- Worked on production backend systems and investigated application behavior across APIs, databases, and service dependencies
- **Tech Stack:** ASP.NET Core, TypeScript, React, React Native, Azure, Entity Framework, Hangfire

PROJECTS

WhatsApp-Integrated Multi-Agent Business Intelligence 2026

- Architected a modular **Multi-Agent System (MAS) with 4 specialized agents** for autonomous strategic task delegation through WhatsApp
- Developed an **Investigative Discovery Loop cross-referencing 10+ API endpoints** to transform backend data into context-aware RAG insights
- Built backend workflows for automated executive reporting from 1000s of data points using a “No Excuses” protocol for comprehensive narrative synthesis
- **Tech Stack:** FastAPI (Python), genai, WhatsApp Business API, Docker, Pydantic, PostgreSQL

Intelligent Legal Multi-Agent RAG 2026-Present

- Architected a modular **Multi-Agent System (MAS) with 4 specialized agents** (Semantic, Entity, Temporal, and Fusion) for autonomous parallelized **Chain-of-Retrieval** reasoning
- Implemented Clean Architecture using FastAPI and Pydantic, achieving 100% architectural decoupling and seamless integration with Gemini
- Developed an **Investigative Discovery Loop** cross-referencing **Qdrant vector embeddings** with **Neo4j** graph entities to transform raw case data into context-aware RAG insights
- Engineered a **Query Decomposer** and **Rewriter** layer to autonomously decompose complex legal mandates into verifiable sub-tasks, increasing retrieval precision
- **Tech Stack:** FastAPI (Python), genai sdk, Qdrant (Vector DB), Neo4j (Graph DB), Redis (Vector Cache), Docker, Pydantic, RQ

Technical Skills

Languages:	Go, Python, C#, TypeScript
Databases:	MySQL, PostgreSQL, Redis, Qdrant, Neo4j, Weaviate
Messaging & Caching:	RabbitMQ, RabbitMQ DLX/DLQ patterns, Redis, Redis Streams, Hangfire, Write-Through Caching, Go Channels
AI Engineering:	LLM Orchestration, Multi-Agent Systems, Agentic Workflows, RAG, Function Calling, MCP, AI Tool Integration, Gemini, OpenCode
Backend Frameworks:	FastAPI, Gin, ASP.NET Core
Infrastructure & Cloud:	Docker, Terraform, GCP, AWS, Cloud Run, GCS, Google Artifact Registry
Observability:	Prometheus, Grafana, Azure AppInsights
Development Practices:	Modular Design, Event-Driven Architecture

AWARDS & ACHIEVEMENTS

2025	**Top 34.5%**	Predicting Road Accident Risk
2021	**Top 10% (185/1747)**	2021 ICPC Asia Dhaka Regional Online Preliminary
2021	**Divisional Champion**	ICPC Asia Dhaka Regional Contest 2021
2022	**Top 12% (206/1648)**	2022 ICPC Asia Dhaka Regional Online Preliminary

EDUCATION

Jatiya Kabi Kazi Nazrul Islam University March 2018 – February 2023
BSc in Computer Science & Engineering Trishal, Mymensingh, Bangladesh